Exact Bézout Certificates and Coefficient Nonvanishing at Section Roots

Lean Stage 2 Algebraic Certificates

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Abstract

We record the Stage 2 Lean formalization of exact rational polynomial reduction and Bézout identities for nine explicit section-coefficient polynomials associated to a fixed degree-24 section modulus, together with the resulting nonvanishing theorems at every root of that modulus, the proof that the modulus has degree 24, the existence of a complex root, and the existence of a complex root at which all nine quantities are simultaneously nonzero. The module `Hodge.ExplicitResidueCertificates` contains 45 theorem declarations; combined with the seven Stage 1 lemmas this yields the audited total of 52 incremental theorems, each depending only on the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. SymPy is used only to generate candidate witnesses; Lean proves every identity independently. No geometric cycle-class, Jacobian-quotient basis, or H/G resultant theorem is claimed.

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1 Scope and non-claims

This note is the companion to the Stage 1 determinant/derivative note. It covers every new Stage 2 theorem in `Hodge/ExplicitResidueCertificates.lean` at commit `e207394a9ed4255ad804860b8f69fde7e940dd54`, organized as one coherent algebraic certificate paper. Generated helper declarations (evaluation lemmas, reduction/Bézout pairs) are mapped as supporting infrastructure for the nine nonvanishing results and three nonvacuity theorems; they are not split into separate papers [1].

Proved here.

- Nine scalar nonvanishing theorems for the formulas w_0, w_1, w_2, w_3, u_0, u_1, u_2, u_3, and $\mathrm{disc} = b^2 - 4c$, under only the field / $\mathbb{Q}$-algebra structure and the modulus-root equation.
- Eighteen exact identities in $\mathbb{Q}[X]$: for each of the nine raw polynomials P, a reduction $P = R + Q \cdot m$ and a Bézout identity $AR + Bm = 1$.
- Degree 24 of the modulus polynomial, existence of a complex root, and simultaneous nonvanishing of all nine quantities at some complex root.

Not proved here.

- The tenth Python certificate: the actual H/G resultant and its common-projective-zero consequence. Pinned mathlib defines resultant via a Sylvester determinant and lists further resultant theory as TODO (`docs/FULL_FORMALIZATION_STAGE2.md`).
- Identification of the U formulas with coefficients in an actual Jacobian quotient (requires a Lean quotient-reduction/basis theorem).
- Complete-intersection ideal regularity, reducedness, saturation; geometric residue-cycle theorems; Betti/Hodge comparison; geometric group action; rational geometric spanning of Hodge classes.
- General or integral Hodge conjecture statements; novelty or peer-review claims.

2 Standing data

Definition 1 (Modulus and section coefficients)**.** Let $m \in \mathbb{Q}[X]$ be the explicit polynomial

$$m = X^{24} + 66081312000\,X^{18} - 4811512860000\,X^{12} + 1167566400000\,X^{6} + 583200000 \quad (1)$$

(`ExplicitResidueCertificates.modulus`). Let $b, c, d, e, g \in \mathbb{Q}[X]$ be the section polynomials `bPoly`,`cPoly`,`dPoly`,`ePoly`,`gPoly`, whose evaluations recover `ExplicitCurve.sb`,...,`sg`. For a

characteristic-zero field K equipped with a $\mathbb{Q}$-algebra structure and $k \in K$, write $b(k) = \mathrm{sb}(k)$ etc., and define the nine scalar formulas

$$w_0 = 3bg - 2ce, \quad w_1 = 6g + be - 4cd, \quad w_2 = 4e - bd - 6c, \qquad w_3 = 2d - 3b, \tag{2}$$

$$u_0 = cw_0 - 6, \qquad u_1 = bw_0 + cw_1, \qquad u_2 = w_0 + bw_1 + cw_2, \quad u_3 = w_1 + bw_2 + ck, \tag{3}$$

$$\mathrm{disc} = b^2 - 4c. \tag{4}$$

The corresponding elements of $\mathbb{Q}[X]$ are `w0Poly`,...,`discPoly`.

Remark 2 (Trust boundary for SymPy)**.** The script `scripts/generate_residue_certificates.py` uses SymPy 1.14.0 [4] to compute candidate sparse rational polynomials R, Q, A, B. Those candidates are written into Lean source; every equality is then proved by Lean polynomial extensionality and exact ring normalization. SymPy is reproducible generation tooling, not part of the proof trust base (`docs/FULL_FORMALIZATION_STAGE2.md`).

3 Supporting polynomial identities

Lemma 3 (Evaluation)**.** *For each section polynomial $P \in \{b, c, d, e, g\}$ and each raw coefficient polynomial $P \in \{w_0, \ldots, w_3, u_0, \ldots, u_3, \mathrm{disc}\}$,*

$$\mathrm{aeval}_k(P_{\mathrm{Poly}}) = P(k) \tag{5}$$

in every characteristic-zero field K equipped with a $\mathbb{Q}$-algebra structure. Lean declares the hypotheses `[Field K]`, `[CharZero K]`, and `[Algebra ℚ K]`. In particular the five section polynomials evaluate to `ExplicitCurve.sb`,...,`sg`.

Lean. Fourteen `@[simp]` theorems (`eval_bPoly` through `eval_disc`) are proved under those field and $\mathbb{Q}$-algebra hypotheses by unfolding definitions and applying the algebra-map simp set. □

Lemma 4 (Reduction and Bézout)**.** *For each of the nine raw polynomials $P \in \mathbb{Q}[X]$ there exist explicitly defined sparse $R, Q, A, B \in \mathbb{Q}[X]$ such that*

$$P = R + Q \cdot m \qquad \text{and} \qquad AR + Bm = 1 \tag{6}$$

as equalities in $\mathbb{Q}[X]$.

Lean. Eighteen theorems `w0_reduction`, `w0_bezout`, ..., `disc_reduction`, `disc_bezout` are proved by `Polynomial.funext`, evaluation at a generic point, and `norm_num`/`ring`. Exact rational coefficients are recorded in the Lean source; they are omitted here only for readability. □

Lemma 5 (Evaluation criterion)**.** *If $P = R + Qm$ and $AR + Bm = 1$ in $\mathbb{Q}[X]$, and if k is a root of m in a characteristic-zero field K equipped with a $\mathbb{Q}$-algebra structure (Lean: `[Field K]`, `[CharZero K]`, `[Algebra ℚ K]`), then* $\mathrm{aeval}_k(P) \neq 0$.

Lean. `eval_ne_zero_of_certificate`, under the same field/$\mathbb{Q}$-algebra hypotheses: evaluate both identities at k; the Bézout identity collapses to $A(k)R(k) = 1$, so $R(k) \neq 0$, hence $P(k) \neq 0$. □

4 Main nonvanishing theorems

Theorem 6 (Nine coefficient nonvanishings)**.** *Let K be a field of characteristic zero with a $\mathbb{Q}$-algebra structure, and let $k \in K$ satisfy*

$$k^{24} + 66081312000\,k^{18} - 4811512860000\,k^{12} + 1167566400000\,k^{6} + 583200000 = 0. \tag{7}$$

Then

$$w_0(k),\ w_1(k),\ w_2(k),\ w_3(k),\ u_0(k),\ u_1(k),\ u_2(k),\ u_3(k),\ \mathrm{disc}(k) \tag{8}$$

are all nonzero.

Lean. Nine theorems `w0_ne_zero`,…,`disc_ne_zero`. Each rewrites the scalar hypothesis as $\mathrm{aeval}_k(m) = 0$ and applies Lemma 5 to the matching reduction/Bézout pair. No numerical root approximation, irreducibility hypothesis, geometric datum, or desired nonvanishing is assumed. □

Theorem 7 (Modulus degree)**.** $\deg(m) = 24$.

Lean. `modulus_degree`: unfold m, apply `compute_degree`, and discharge closed numerical bounds by `norm_num` and ordinary `decide +kernel`. This is not a `native_decide` shortcut on the algebraic content of the nonvanishing theorems. □

Theorem 8 (Complex root existence)**.** *There exists $k \in \mathbb{C}$ with* $\mathrm{aeval}_k(m) = 0$.

Lean. `exists_complex_root`: apply algebraic closedness of $\mathbb{C}$ (`IsAlgClosed.exists_aeval_eq_zero`) using Theorem 7 and $\deg m \neq 0$. □

Theorem 9 (Simultaneous nonvanishing at a complex root)**.** *There exists $k \in \mathbb{C}$ satisfying the modulus equation such that*

$$w_0(k) \neq 0,\ \ldots,\ w_3(k) \neq 0,\ u_0(k) \neq 0,\ \ldots,\ u_3(k) \neq 0,\ \mathrm{disc}(k) \neq 0. \tag{9}$$

Lean. `exists_complex_nonvanishing`: take a root from Theorem 8 and apply all nine cases of Theorem 6. This shows the universal nonvanishing statements are not vacuous assertions over a field without section roots. It is not a geometric cycle-class assertion. □

5 Theorem–declaration mapping

Main results

#	Lean declaration	Role
1	`w0_ne_zero`	$w_0 = 3bg - 2ce \neq 0$
2	`w1_ne_zero`	$w_1 = 6g + be - 4cd \neq 0$
3	`w2_ne_zero`	$w_2 = 4e - bd - 6c \neq 0$
4	`w3_ne_zero`	$w_3 = 2d - 3b \neq 0$
5	`u0_ne_zero`	$u_0 = cw_0 - 6 \neq 0$
6	`u1_ne_zero`	$u_1 = bw_0 + cw_1 \neq 0$
7	`u2_ne_zero`	$u_2 = w_0 + bw_1 + cw_2 \neq 0$
8	`u3_ne_zero`	$u_3 = w_1 + bw_2 + ck \neq 0$

9	`disc_ne_zero`	$b^2 - 4c \neq 0$
10	`modulus_degree`	$\deg m = 24$
11	`exists_complex_root`	$\exists\, k \in \mathbb{C},\ m(k) = 0$
12	`exists_complex_nonvanishing`	simultaneous nonvanishing

Supporting certificates (same paper; not split)

Count	Lean declarations	Role
5	`eval_bPoly...eval_gPoly`	section evaluation
9	`eval_w0...eval_disc`	raw-coefficient evaluation
9	`*_reduction`	$P = R + Qm$ in $\mathbb{Q}[X]$
9	`*_bezout`	$AR + Bm = 1$ in $\mathbb{Q}[X]$
1	`eval_ne_zero_of_certificate`	evaluation criterion

Total Stage 2 theorems: $12+5+9+9+9+1 = 45$. Combined with the seven Stage 1 lemmas: 52 audited declarations (`ResidueAxiomAudit.lean`).

6 Axiom audit and verification evidence

The checker `scripts/check_residue_axioms.py` requires `ResidueAxiomAudit.lean` to enumerate every theorem in both incremental modules in source order, builds `Hodge.ExplicitResidueAlgebra` and `Hodge.ExplicitResidueCertificates`, and accepts only `propext`, `Classical.choice`, `Quot.sound`.

Verified log lines (America/Chicago evidence timestamps on disk):

- `build-and-audit-final.log`: reports successful `Hodge.ExplicitResidueCertificates` build (7085/7085) and "PASS: 52 incremental theorem declarations; standard axioms only".
- `residue-audit-e207394.log`: same PASS line; 52 axiom-print entries.
- `no-sorry-e207394.log`: "OK: no sorry/admit/native_decide, no axiom declarations".
- `full-build-e207394.log`: full repository `lake build` completed successfully at commit `e207394`.
- Prior regeneration check (certificate delivery evidence): byte-identical SymPy regeneration of `ExplicitResidueCertificates.lean`; `LICENSE`, `CITATION.cff`, and Stage 1 Lean source unchanged.

7 Toolchain pins and reproducibility

Pin	Value
Lean [2]	`leanprover/lean4:v4.22.0`
mathlib input [3]	`v4.22.0`
mathlib checkout	`79e94a093aff4a60fb1b1f92d9681e407124c2ca`
SymPy (generator only) [4]	1.14.0
Git commit	`e207394a9ed4255ad804860b8f69fde7e940dd54`
PR	https://github.com/BenFrohman/HODGE/pull/36
License	Apache-2.0 (preserved; no relicense)

```
python3 scripts/generate_residue_certificates.py
# optional: git diff --exit-code -- Hodge/ExplicitResidueCertificates.lean
lake build Hodge.ExplicitResidueAlgebra \
           Hodge.ExplicitResidueCertificates
python3 scripts/check_residue_axioms.py
python3 scripts/check_no_sorry.py
```

Local sources and evidence inspected (read-only; no repository edits). Mathematical source pin `e207394a9ed4255ad804860b8f69fde7e940dd54`; later inspected worktree `fefe2ba9d87c900b538d02c13f7f51f375fcd285` retains identical reviewed modules, audit, and toolchain/lock files:

- `/workspace/hodge-formal-stage1-20261106/Hodge/ExplicitResidueCertificates.lean`
- `/workspace/hodge-formal-stage1-20261106/ResidueAxiomAudit.lean`
- `/workspace/hodge-formal-stage1-20261106/docs/FULL_FORMALIZATION_STAGE2.md`
- `/workspace/hodge-formal-stage1-20261106/scripts/generate_residue_certificates.py`
- `/workspace/hodge-formal-stage2-evidence-20261106/` (actual files: `build-and-audit-final.log`, `build-and-audit.log`, `full-build.log`, `typecheck*.log`, `DegreeCheck.lean`, `generate_certificates.py`)
- `/workspace/hodge-formal-stage1-evidence-20261106/` (`Stage1Audit.lean`, `axioms.txt`, `baseline-explicit-curve.lean`)
- `/workspace/hodge-certificate-delivery-20261106/` (`full-build-e207394.log`, `residue-audit-e207394.log`, `no-sorry-e207394.log`, status JSON).

CI verification at a later identical head. User-reported merge of PR https://github.com/BenFrohman/HODGE/pull/36 is acknowledged; this packaging records no merge-commit SHA (merge verification is underway separately). At head `fefe2ba9d87c900b538d02c13f7f51f375fcd285`, where the reviewed Lean modules, audit, and toolchain/lock match mathematical pin `e207394a9ed4255ad804860b8f69fde7e940dd54`, all five reported GitHub Actions checks completed successfully: Lean/build and Lean/attempt (https://github.com/BenFrohman/HODGE/actions/runs/42317080169),

Lean certificate/certify on the PR (https://github.com/BenFrohman/HODGE/actions/runs/42317080616), Lean certificate/certify on the push (https://github.com/BenFrohman/HODGE/actions/runs/42317075046), and Paper/paper (https://github.com/BenFrohman/HODGE/actions/runs/42317080271). Both certification logs build the incremental modules and print `PASS: 52 incremental theorem declarations; standard axioms only`. CI success verifies those algebraic declarations; it does not discharge geometric Hodge/Betti/cycle-class obligations. This note remains a REVIEW DRAFT with no DOI and no publication claim.

8 Limitations relative to the geometric program

The nine nonvanishings are algebraic statements about evaluations of explicit elements of $\mathbb{Q}[X]$ at roots of m. They do not by themselves produce geometric residues, nor do they prove that the corresponding Jacobian-quotient classes are nonzero in a geometric cohomology theory. Villaflor Loyola's complete-intersection residue theorem [5] and Griffiths' comparison remain external classical input (`NOTICE.md`).

The classical Python projection certificate checks ten exact unit-at-every-root identities (eight W/U formulas, the discriminant, and the H/G resultant). Lean Stage 2 covers the first nine; the tenth (resultant) remains unformalized.

These algebraic certificates can stand as fully formalized Lean mathematics while a full geometric Hodge paper for the sextic fourfold remains unpublished pending the missing geometric foundations. This REVIEW DRAFT assigns no DOI and makes no upload.

Author, license, and contact

Author metadata: **Frohman, Benjamin Stanley**. Email: `frohmanbenjamin@gmail.com`. ORCID: 0009-0006-7068-3718. Verified from repository `CITATION.cff`, `NOTICE`, and `.zenodo.json`. Copyright (c) 2026 Benjamin Stanley Frohman. Apache License 2.0. Historical archive DOI `10.5281/zenodo.23629480` is provenance of prior HODGE releases, not a DOI for this Stage 2 note.

References

[1] Benjamin Stanley Frohman. HODGE: Lean 4 formalization and hodge ledger. https://github.com/BenFrohman/HODGE, 2026. Source commit `e207394a9ed4255ad804860b8f69fde7e940dd54`; PR https://github.com/BenFrohman/HODGE/pull/36. Historical archive DOI `10.5281/zenodo.23629480` identifies prior repository provenance and is not a DOI for these incremental algebra notes.

[2] Lean FRO. The Lean 4 theorem prover. https://github.com/leanprover/lean4, 2026. Pinned release `v4.22.0` via `lean-toolchain`.

[3] mathlib community. mathlib4. https://github.com/leanprover-community/mathlib4, 2026. Pinned input revision `v4.22.0`; locked checkout `79e94a093aff4a60fb1b1f92d9681e407124c2ca` from `lake-manifest.json`.

[4] Aaron Meurer et al. SymPy: symbolic computing in Python, 2026. Used only to generate candidate rational witnesses; not part of the Lean proof trust base.

[5] Roberto Villaflor Loyola. Periods of complete intersection algebraic cycles. *Manuscripta Mathematica*, 167:765–792, 2022. Cited as classical residue-cycle input; not formalized in the incremental Lean modules of this note.